Housing Prices Analysis

This analysis uses a dataset relative to houses prices in the USA. The objective is to analyse which are the factors that contribute the most to the prices of houses.

First of all I import libraries, import the data and visualize it

```r
library(dplyr)
library(ggplot2)
df<-read.csv("C:/Users/s1720/OneDrive/Desktop/GitHub/DataSet/Housing.csv")
str(df) #Checking data structure
head(df) #Importing the first values of each column
```

```
'data.frame':   545 obs. of  13 variables:
 $ price       : int  13300000 12250000 12250000 12215000 11410000 10850000 10150000 10150000 98700
 $ area            : int  7420 8960 9960 7500 7420 7500 8580 16200 8100 5750 ...
 $ bedrooms        : int  4 4 3 4 4 3 4 5 4 3 ...
 $ bathrooms       : int  2 4 2 2 1 3 3 3 1 2 ...
 $ stories         : int  3 4 2 2 2 1 4 2 2 4 ...
 $ mainroad        : chr  "yes" "yes" "yes" "yes" ...
 $ guestroom       : chr  "no" "no" "no" "no" ...
 $ basement        : chr  "no" "no" "yes" "yes" ...
 $ hotwaterheating : chr  "no" "no" "no" "no" ...
 $ airconditioning : chr  "yes" "yes" "no" "yes" ...
 $ parking         : int  2 3 2 3 2 2 2 0 2 1 ...
 $ prefarea        : chr  "yes" "no" "yes" "yes" ...
 $ furnishingstatus: chr  "furnished" "furnished" "semi-furnished" "furnished" ...
     price area bedrooms bathrooms stories mainroad guestroom basement
1 13300000 7420        4         2       3      yes        no       no
2 12250000 8960        4         4       4      yes        no       no
3 12250000 9960        3         2       2      yes        no      yes
4 12215000 7500        4         2       2      yes        no      yes
5 11410000 7420        4         1       2      yes       yes      yes
6 10850000 7500        3         3       1      yes        no      yes
  hotwaterheating airconditioning parking prefarea furnishingstatus
1              no             yes       2      yes        furnished
2              no             yes       3       no        furnished
3              no              no       2      yes   semi-furnished
4              no             yes       3      yes        furnished
5              no             yes       2       no        furnished
6              no             yes       2      yes   semi-furnished
```

Checking if the dataset needs cleaning

```r
mean(df$price)#If there are non numerical values we get NA
mean(df$area)
mean(df$bedrooms)
mean(df$bathrooms)
mean(df$stories)
mean(df$parking)
sum(is.na(df)) #Checking if there are NA values in general including char columns
sapply(df,length) #Checking that the values have the same length
df <- df[!duplicated(df), ]  #Eliminate duplicates if present
```

```
[1] 4766729
[1] 5150.541
[1] 2.965138
```

```
[1] 1.286239
[1] 1.805505
[1] 0.693578
[1] 0
           price             area         bedrooms        bathrooms 
             545              545              545              545 
         stories         mainroad        guestroom         basement 
             545              545              545              545 
 hotwaterheating  airconditioning          parking         prefarea 
             545              545              545              545 
furnishingstatus 
             545 
```

Since there needs to be no cleaning I go on and convert the yes and no values into 0 and 1 dummy variables (0, 1 and 2 in the case of furniture)

```r
#First the variables with yes and no
binary_cols <- c("mainroad", "guestroom", "basement", "hotwaterheating",
                 "airconditioning", "prefarea")
df[binary_cols]<-lapply(df[binary_cols],function(x) ifelse(x=="yes",1,0))
#Second the furniture variable
df[["furnishingstatus"]]<-lapply(df[["furnishingstatus"]],function(x) ifelse(x=="furnished",2,i
#str(df)
df$furnishingstatus <- unlist(df$furnishingstatus) #Changing the column from list to num
str(df)
```

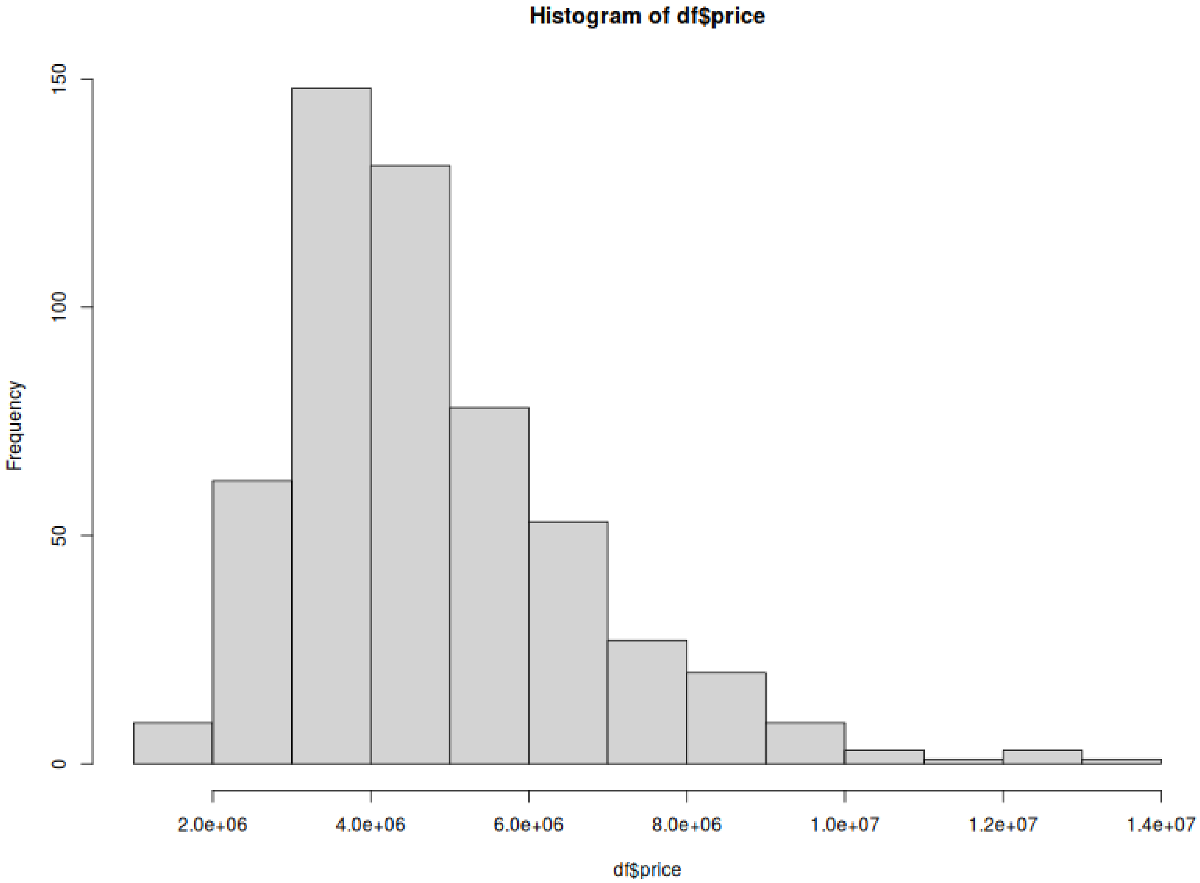
```
'data.frame':   545 obs. of  13 variables:
 $ price       : int  13300000 12250000 12250000 12215000 11410000 10850000 10150000 10150000 98700
 $ area            : int  7420 8960 9960 7500 7420 7500 8580 16200 8100 5750 ...
 $ bedrooms        : int  4 4 3 4 4 3 4 5 4 3 ...
 $ bathrooms       : int  2 4 2 2 1 3 3 3 1 2 ...
 $ stories         : int  3 4 2 2 2 1 4 2 2 4 ...
 $ mainroad        : num  1 1 1 1 1 1 1 1 1 1 ...
 $ guestroom       : num  0 0 0 0 1 0 0 0 1 1 ...
 $ basement        : num  0 0 1 1 1 1 0 0 1 0 ...
 $ hotwaterheating : num  0 0 0 0 0 0 0 0 0 0 ...
 $ airconditioning : num  1 1 0 1 1 1 1 0 1 1 ...
 $ parking         : int  2 3 2 3 2 2 2 0 2 1 ...
 $ prefarea        : num  1 0 1 1 0 1 1 0 1 1 ...
 $ furnishingstatus: num  2 2 1 2 2 1 1 0 2 0 ...
```

Then I calculate summary statistics and visualize a histogram of the data

```r
summary(df[,c("price","area","bedrooms","bathrooms","stories")])
hist(df$price)
```

```
     price               area          bedrooms       bathrooms    
 Min.   : 1750000   Min.   : 1650   Min.   :1.000   Min.   :1.000  
 1st Qu.: 3430000   1st Qu.: 3600   1st Qu.:2.000   1st Qu.:1.000  
 Median : 4340000   Median : 4600   Median :3.000   Median :1.000  
 Mean   : 4766729   Mean   : 5151   Mean   :2.965   Mean   :1.286  
 3rd Qu.: 5740000   3rd Qu.: 6360   3rd Qu.:3.000   3rd Qu.:2.000  
 Max.   :13300000   Max.   :16200   Max.   :6.000   Max.   :4.000  
    stories     
 Min.   :1.000  
 1st Qu.:1.000  
```

```
 Median :2.000  
 Mean   :1.806  
 3rd Qu.:2.000  
 Max.   :4.000  
```



As we can see the prices of the houses go from 1.750.000,00 to 13.300.000,00$, with a median of 4.340.000,00 dollars. The area has a minimum of 1.650 sq. ft. and a maximum of 16.200, the median is 4600. In both cases the median is smaller than the mean suggesting the presence of a skewness to the left and of some values extremely high values that noticeably increase the mean.

Now there I regress the prices on all the other variables

```r
model<-lm(price~ area+bedrooms+bathrooms+stories+mainroad+guestroom+basement+hotwaterheating+ai
summary(model)
```

```
Call:
lm(formula = price ~ area + bedrooms + bathrooms + stories + 
    mainroad + guestroom + basement + hotwaterheating + airconditioning + 
    parking + prefarea + furnishingstatus, data = df)

Residuals:
     Min       1Q   Median       3Q      Max 
-2657636  -642433   -47099   508148  5098135 

Coefficients:
```

```
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)      -324509.05  236816.49  -1.370 0.171172
area                 243.91      24.33  10.024  < 2e-16 ***
bedrooms          119474.39   72675.18   1.644 0.100777
bathrooms         988888.29  103543.07   9.551  < 2e-16 ***
stories           450391.52   64282.34   7.006 7.42e-12 ***
mainroad          423100.74  142472.67   2.970 0.003115 **
guestroom         298030.51  131935.97   2.259 0.024293 *
basement          357926.36  110383.85   3.243 0.001259 **
hotwaterheating   872936.03  223311.48   3.909 0.000105 ***
airconditioning   853633.59  108341.74   7.879 1.87e-14 ***
parking           279785.64   58608.60   4.774 2.34e-06 ***
prefarea          647055.60  115857.64   5.585 3.73e-08 ***
furnishingstatus  213187.78   63059.84   3.381 0.000776 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1070000 on 532 degrees of freedom
Multiple R-squared:  0.6801,    Adjusted R-squared:  0.6729
F-statistic: 94.24 on 12 and 532 DF,  p-value: < 2.2e-16
```

All variables, aside from the number of bedrooms, are significantly different from 0 at the 5% significance level. Now I eliminate the number of bedrooms.

```r
model<-lm(price~ area+bathrooms+stories+mainroad+guestroom+basement+hotwaterheating+airconditio
summary(model)
```

```
Call:
lm(formula = price ~ area + bathrooms + stories + mainroad +
    guestroom + basement + hotwaterheating + airconditioning +
    parking + prefarea + furnishingstatus, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-2731863  -656171   -86893   474068  5141007

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)       -94631.42  191430.45  -0.494 0.621271
area                 247.06      24.29  10.169  < 2e-16 ***
bathrooms        1026884.35  101091.60  10.158  < 2e-16 ***
stories           487539.50   60275.81   8.088 4.10e-15 ***
mainroad          394465.18  141629.51   2.785 0.005540 **
guestroom         293050.50  132111.68   2.218 0.026962 *
basement          383161.17  109485.79   3.500 0.000505 ***
hotwaterheating   880183.65  223624.27   3.936 9.38e-05 ***
airconditioning   851457.56  108506.54   7.847 2.35e-14 ***
parking           286628.53   58553.90   4.895 1.30e-06 ***
prefarea          650894.81  116018.96   5.610 3.25e-08 ***
furnishingstatus  216987.70   63118.03   3.438 0.000632 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1072000 on 533 degrees of freedom
Multiple R-squared:  0.6784,    Adjusted R-squared:  0.6718
```

```
F-statistic: 102.2 on 11 and 533 DF,  p-value: < 2.2e-16
```

Now all explanatory variables of the model are significantly different from zero at the 0.05 significance level. In this model we see the highest effect on price is given by the number of bathrooms, the presence of air conditioning, of hot water heating, the presence in a preferential area and the presence of an access to a mainroad. The R-squared is quite high suggesting a good capacity of the model to explain the prices of housing in America.

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Source of the Dataset: https://www.kaggle.com/datasets/yasserh/housing-prices-dataset/data